IPsec Workshop

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Dresden

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Avoid frame copy in skb_cow_data

- **Problem:** Most of the ESP data frames are linearized with skb_cow_data.
- **Solved TX:** Use separate src and dst buffers for crypto operations, RFC code exists.
- **Solved RX:** Linearize only if the buffer is not writable, RFC code exists.
- **New Problem:** Some crypto algorithm implementations linearize if a nonlinear buffer is passed in (gcm-aesni).

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- **General forwarding problem:** Such buffers can't be offloaded to hardware, we need to linearize and segment them in the stack.
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- **GSO:** Works if we can offload the crypto operations to a NIC, RFC code exists.
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Some performance numbers

- Tunnel mode forwarding with rfc4106-gcm-aesni-256 crypto and IPv4 TCP.
- 2 IPsec gateways: Intel(R) Xeon(R) CPU E7-8890 v3 @ 2.50GHz (72 cores), 16 x 10 Gbps.
- Traffic generator: Spirent Testcenter.
- **Baseline (single flow):** 3.8 Gbps at layer 1 (April 2016).
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